

Volatility Forecasting Using Machine Learning: A Comparative Study of Econometric and Machine Learning Models

Abstract

Volatility forecasting plays a critical role in financial risk management, portfolio optimization, derivative pricing, and algorithmic trading. This study investigates whether machine learning techniques can improve volatility forecasting performance relative to traditional econometric models. Using daily market data from Apple Inc. (AAPL), the SPDR S&P 500 ETF (SPY), Microsoft Corporation (MSFT), and the CBOE Volatility Index (VIX) from 2016 to 2025, a comprehensive feature set was constructed incorporating lagged returns, realized volatility measures, market sentiment indicators, and trading activity variables. Realized volatility was estimated using the Garman-Klass estimator, and next-day volatility forecasting was formulated as a supervised learning problem. Baseline forecasting models were first established to provide benchmark performance. The study subsequently evaluates econometric approaches such as GARCH and EGARCH and machine learning methods including Random Forest and XGBoost. The objective is to determine whether machine learning models can deliver superior predictive accuracy while capturing the nonlinear dynamics of financial market volatility.

1. Introduction

Financial market volatility represents the degree of variation in asset prices over time and serves as a fundamental measure of risk in modern finance. Accurate volatility forecasts are essential for portfolio management, risk assessment, derivatives pricing, asset allocation, and regulatory compliance. As financial markets become increasingly complex and interconnected, the demand for reliable volatility forecasting techniques continues to grow.

One of the most widely observed characteristics of financial volatility is volatility clustering, whereby periods of high volatility tend to be followed by additional periods of elevated volatility, while tranquil periods are often followed by continued market stability. Traditional econometric models such as the Generalized Autoregressive Conditional Heteroskedasticity (GARCH) model were specifically developed to capture these characteristics and have become standard tools in financial forecasting.

Recent developments in machine learning have introduced alternative approaches capable of modeling nonlinear relationships, complex feature interactions, and large multidimensional datasets. Methods such as Random Forests and Gradient Boosting Machines have demonstrated promising results across numerous forecasting applications. However, their effectiveness in volatility forecasting remains an active area of research.

This study compares traditional econometric techniques and modern machine learning approaches within a unified forecasting framework. By evaluating each model on identical data and using consistent evaluation metrics, the study aims to provide empirical evidence regarding the relative strengths and weaknesses of these competing methodologies.

Research Question

Can machine learning models outperform traditional econometric models in forecasting next-day equity volatility?

Objectives

- Construct a comprehensive volatility forecasting dataset.
- Develop benchmark forecasting models.
- Evaluate GARCH and EGARCH volatility models.
- Evaluate Random Forest and XGBoost models.
- Compare forecasting performance across all model classes.
- Identify the most informative predictors of future volatility.

2. Data Description

Data Source

Historical market data were obtained from Yahoo Finance using the Python `yfinance` library.

Sample Period

The study covers the period from January 2016 through December 2025.

Assets Included

- Apple Inc. (AAPL)
- SPDR S&P 500 ETF (SPY)
- Microsoft Corporation (MSFT)
- CBOE Volatility Index (VIX)

Target Variable

The forecasting target is next-day realized volatility of Apple Inc. (AAPL), estimated using the Garman-Klass volatility estimator.

Feature Categories

The final dataset contains multiple groups of predictive variables:

Return Features

- Daily AAPL returns
- One-day lagged returns
- Five-day lagged returns
- Ten-day lagged returns

Volatility Features

- Five-day rolling volatility
- Twenty-day rolling volatility
- Sixty-day rolling volatility
- Volatility momentum

Market Sentiment Features

- VIX closing level
- Daily VIX change

Market-Wide Features

- SPY daily return
- SPY realized volatility

Cross-Asset Features

- MSFT daily return
- MSFT realized volatility

Trading Activity Features

- AAPL trading volume
- Daily volume change

Following feature construction and data cleaning, the final modeling dataset consisted of eighteen variables suitable for predictive modeling.

3. Methodology

Volatility Estimation

Realized volatility was estimated using the Garman-Klass estimator, which incorporates information from open, high, low, and close prices. Compared with simple close-to-

close volatility measures, the Garman-Klass estimator utilizes intraday price movements and is generally considered more efficient for volatility estimation.

Feature Engineering

Features were constructed to capture multiple dimensions of market behavior, including return dynamics, volatility persistence, market sentiment, cross-asset relationships, and trading activity. Lagged variables and rolling volatility measures were included to account for temporal dependencies commonly observed in financial time series.

Forecasting Framework

To prevent look-ahead bias, all forecasts were generated using chronological train-test splits. The first portion of the dataset was treated as the training sample, while subsequent observations were reserved for out-of-sample evaluation. This approach more closely resembles real-world forecasting conditions.

Evaluation Metrics

Forecast performance is assessed using:

- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- Mean Absolute Error (MAE)
- QLIKE Loss

QLIKE is included because it is widely regarded as one of the most appropriate evaluation metrics for volatility forecasting models.

4. Baseline Models

Before evaluating advanced forecasting approaches, two benchmark models were constructed.

4.1 Naive Forecast

The naive forecasting model assumes that tomorrow's volatility will equal today's realized volatility. Due to the persistence typically observed in volatility series, this benchmark often provides a surprisingly strong reference point.

4.2 Historical Average Forecast

The historical average model forecasts future volatility using the twenty-day rolling average of realized volatility. This benchmark reflects the assumption that volatility reverts toward recent historical levels.

4.3 Baseline Results

Model	QLIKE	RMSE	MAE
Naive Forecast	0.4451	0.5997	0.4363
Historical Average Forecast	0.3385	0.5819	0.3927

The historical average model outperformed the naive forecast across all evaluation metrics. The largest improvement occurred in QLIKE loss, which decreased from 0.4451 to 0.3385. Similar improvements were observed in RMSE and MAE. These results suggest that recent average volatility contains more predictive information about future volatility than a simple persistence assumption. Consequently, the historical average model serves as the primary benchmark for all subsequent forecasting models.